

Quantium Virtual Internship

Retail Strategy and Analytics — Task 2

Trial Store Evaluation: Stores 77, 86 and 88

Prepared for: Julia (Category Manager) & Zilinka

Dataset: QVI_data.csv (Task 1 output)

Analysis conducted in Python (pandas, scipy, matplotlib)

1. Introduction & Objective

A store trial was run in stores 77, 86 and 88 between February and April 2019. Julia wants to know whether the trial drove a measurable uplift in chip sales, and if so, whether it was driven by more customers, more visits per customer, or bigger baskets. To isolate the trial's effect from normal seasonal and store-level variation, we match each trial store to a similar "control" store that did not receive the trial, then compare trial vs. control performance before and during the trial period.

2. Building Monthly Store Metrics

```
data = pd.read_csv('QVI_data.csv')
data['DATE'] = pd.to_datetime(data['DATE'])
data['YEARMONTH'] = data['DATE'].dt.year * 100 + data['DATE'].dt.month

measureOverTime = data.groupby(['STORE_NBR', 'YEARMONTH']).apply(lambda g: pd.Series({
    'totSales':      g['TOT_SALES'].sum(),
    'nCustomers':    g['LYLTY_CARD_NBR'].nunique(),
    'nTxnPerCust':    g['TXN_ID'].nunique() / g['LYLTY_CARD_NBR'].nunique(),
    'nChipsPerTxn':   g['PROD_QTY'].sum() / g['TXN_ID'].nunique(),
    'avgPricePerUnit': g['TOT_SALES'].sum() / g['PROD_QTY'].sum(),
})).reset_index()

# Keep only stores with a full 12 months of data, and only the pre-trial window
storesWithFullObs = measureOverTime.groupby('STORE_NBR').size()
storesWithFullObs = storesWithFullObs[storesWithFullObs == 12].index
preTrialMeasures = measureOverTime[(measureOverTime['YEARMONTH'] < 201902) &
                                     (measureOverTime['STORE_NBR'].isin(storesWithFullObs))]
```

For each store and month we calculate total sales, number of distinct customers, transactions per customer, chips (units) per transaction, and average price per unit. 260 of the 272 stores have a complete 12-month observation history and are kept as potential control stores.

3. Selecting Control Stores

For each trial store, every other fully-observed store is scored on how closely its **pre-trial** monthly total sales and customer counts tracked the trial store, using two complementary measures: a Pearson correlation (how similar the trend/shape is) and a standardised magnitude distance (how similar the absolute level is, scaled 0–1). The two are averaged into a single score per driver, then averaged again across the sales and customer drivers to get one overall similarity score per candidate store.

```
def calculateCorrelation(inputTable, metricCol, storeComparison):
    trial_series = inputTable[inputTable.STORE_NBR == storeComparison].set_index('YEARMONTH')[metricCol]
    rows = []
    for i in inputTable['STORE_NBR'].unique():
        other_series = inputTable[inputTable.STORE_NBR == i].set_index('YEARMONTH')[metricCol]
        joined = pd.concat([trial_series, other_series], axis=1, join='inner')
        corr = joined.iloc[:,0].corr(joined.iloc[:,1])
        rows.append({'Store1': storeComparison, 'Store2': i, 'corr_measure': corr})
    return pd.DataFrame(rows)

def calculateMagnitudeDistance(inputTable, metricCol, storeComparison):
    # abs difference per month, standardised to 0-1 (1 = most similar), then averaged across months
    ... # see full code in appendix; mirrors the corr function's loop structure

def select_control_store(preTrialMeasures, trial_store, corr_weight=0.5):
    corr_nSales = calculateCorrelation(preTrialMeasures, 'totSales', trial_store)
    corr_nCustomers = calculateCorrelation(preTrialMeasures, 'nCustomers', trial_store)
    magnitude_nSales = calculateMagnitudeDistance(preTrialMeasures, 'totSales', trial_store)
    magnitude_nCustomers = calculateMagnitudeDistance(preTrialMeasures, 'nCustomers', trial_store)

    scoreNSales = corr_weight * corr_nSales + (1 - corr_weight) * magnitude_nSales
```

```

scoreNCust = corr_weight * corr_nCustomers + (1 - corr_weight) * magnitude_nCustomers
finalControlScore = 0.5 * scoreNSales + 0.5 * scoreNCust
# control store = highest-scoring store that isn't the trial store itself
return finalControlScore.sort_values(ascending=False)

```

3.1 Results

Trial store 77 → control store **233**

Candidate store	Sales score	Customer score	Final score
233	0.945	0.992	0.968
41	0.874	0.909	0.892
17	0.862	0.855	0.858

Trial store 86 → control store **155**

Candidate store	Sales score	Customer score	Final score
155	0.920	0.964	0.942
109	0.875	0.868	0.872
114	0.827	0.895	0.861

Trial store 88 → control store **237**

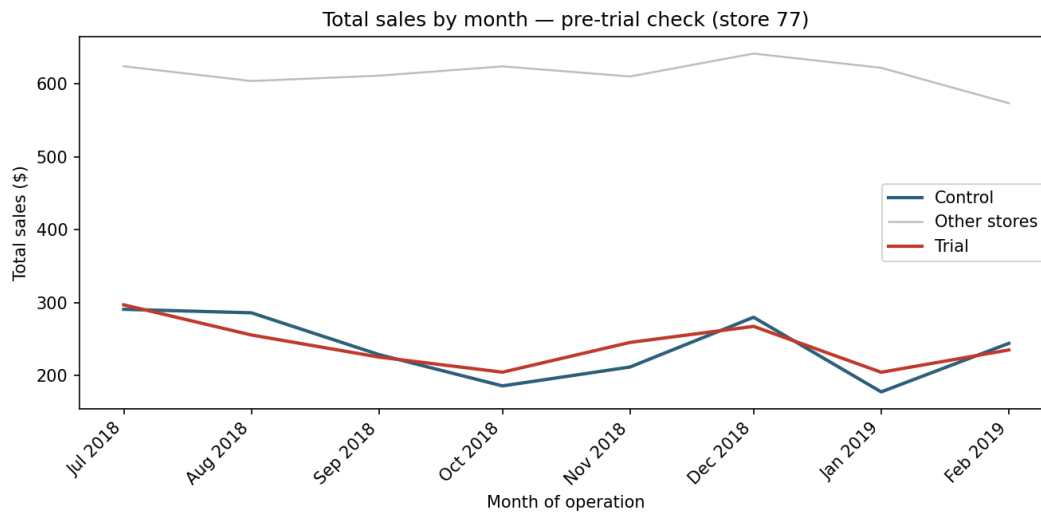
Candidate store	Sales score	Customer score	Final score
237	0.632	0.967	0.800
178	0.714	0.880	0.797
69	0.579	0.842	0.711

Store 233 is selected for trial store 77, store 155 for trial store 86, and store 237 for trial store 88 — each is the highest-scoring, non-trial store on the combined sales + customer similarity measure.

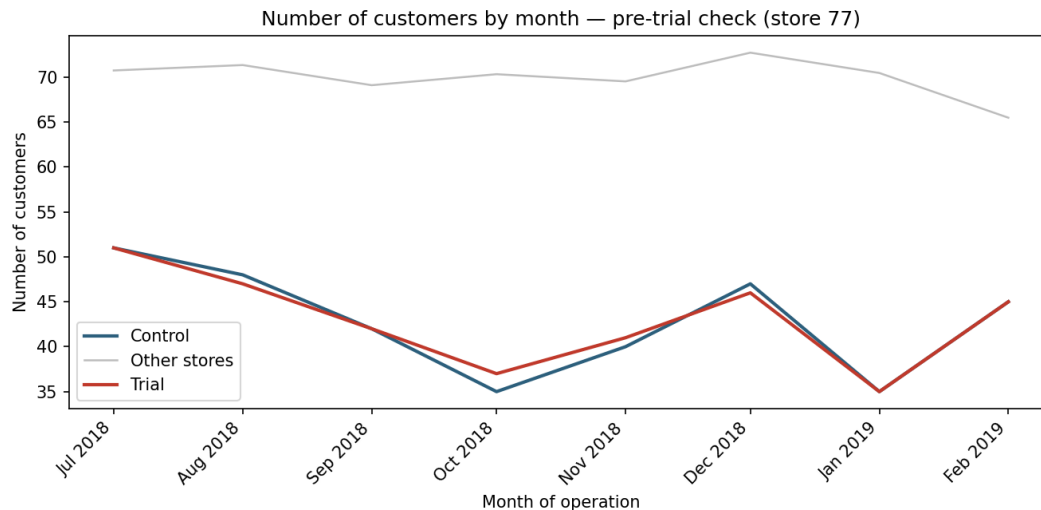
4. Trial Store 77

4.1 Pre-trial visual check

Before assessing the trial, we confirm trial store 77 and control store 233 moved together in the eight months before the trial (Jul 2018 – Feb 2019).



Pre-trial monthly total sales — store 77 vs control 233 vs other stores.



Pre-trial monthly customer counts — store 77 vs control 233 vs other stores.

The trial and control lines track each other closely prior to the trial, confirming this is a suitable control store.

4.2 Trial impact — total sales

```
scalingFactor = preTrial[trial].totSales.sum() / preTrial[control].totSales.sum()
scaledControlSales = control.totSales * scalingFactor
percentageDiff = abs(scaledControlSales - trial.totSales) / scaledControlSales
stdDev = percentageDiff[pre-trial months].std()
tValue = percentageDiff / stdDev # vs t-critical at 95th pct, df=7
```

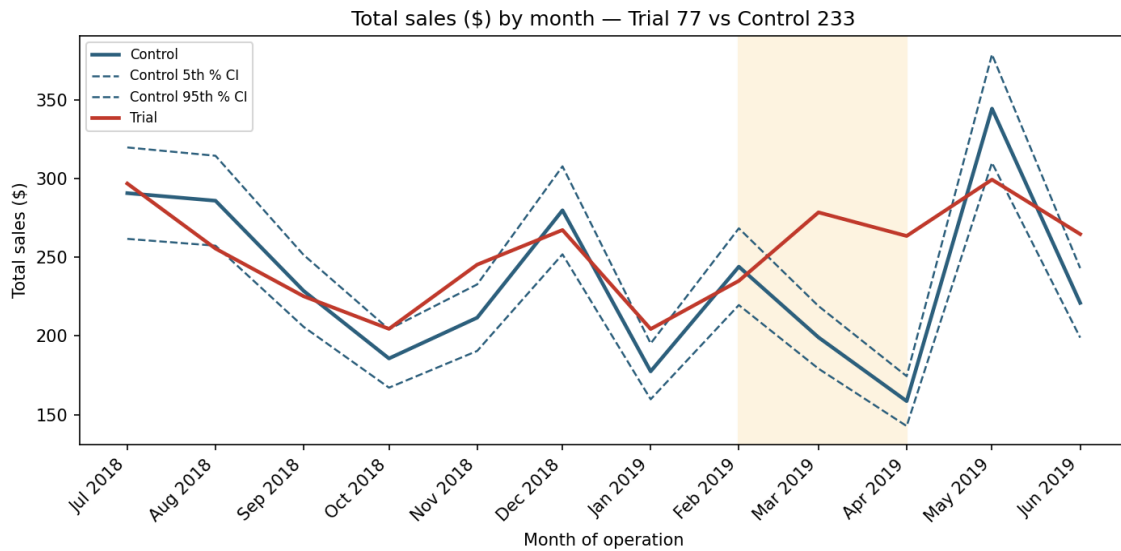


Figure: Trial vs control total sales with control store's 5th–95th percentile band (shaded = trial period).

Result: scaling factor = 1.024, pre-trial standard deviation of scaled %-difference = 0.050, t-critical (95th pct, df=7) = 1.89. Statistically significant sales uplift months: **03/2019, 04/2019**.

4.3 Trial impact — number of customers

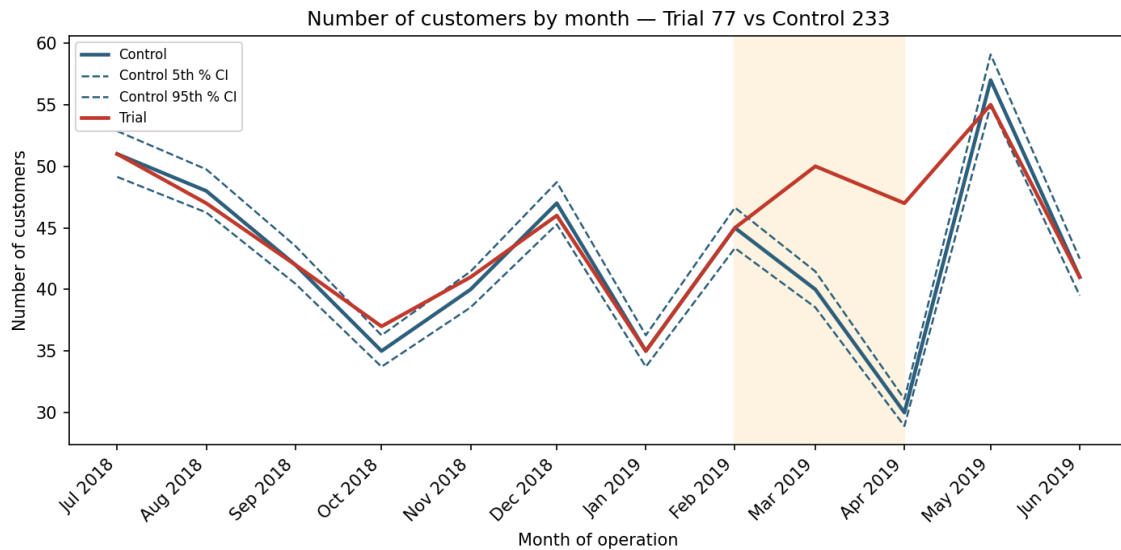


Figure: Trial vs control number of customers with control store's 5th–95th percentile band.

Result: statistically significant customer-count uplift months: **03/2019, 04/2019**.

4.4 What drove the change?

Store	Sales	Customers	Txn/Cust	Units/Txn	Price/Unit
Trial	6.7%	10.8%	-1.7%	3.6%	-5.8%
Control	-15.4%	-10.0%	0.4%	-4.8%	-1.3%

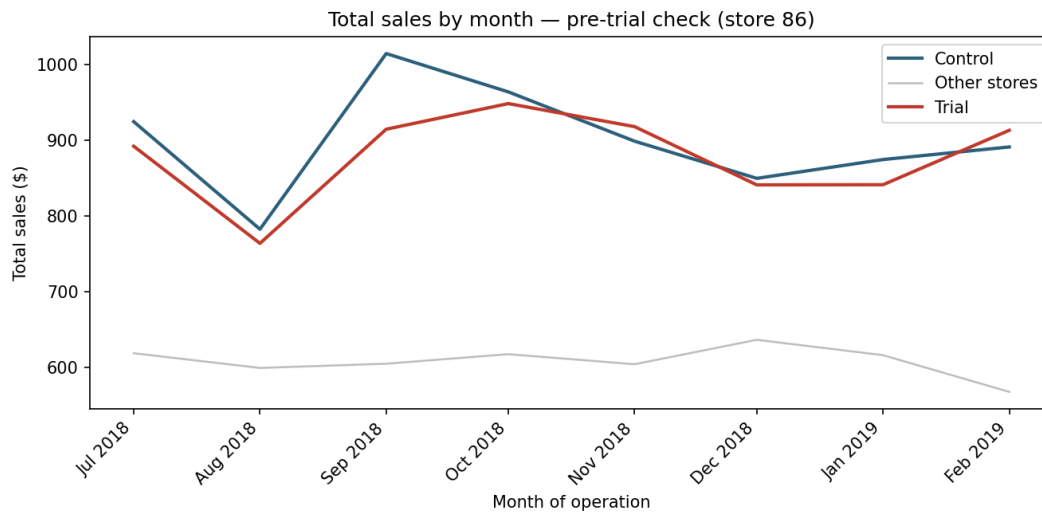
Average % change, trial period (Feb–Apr 2019) vs. pre-trial monthly average.

While control store 233's sales fell 15.4% over the same window (in line with the wider seasonal dip seen across most stores after the December peak), trial store 77's sales rose 6.7%. The gap is explained mainly by **more customers shopping at the store, rather than existing customers buying more per visit** (10.8% more customers vs. -1.7% change in visits/customer and 3.6% change in units/visit for the trial store).

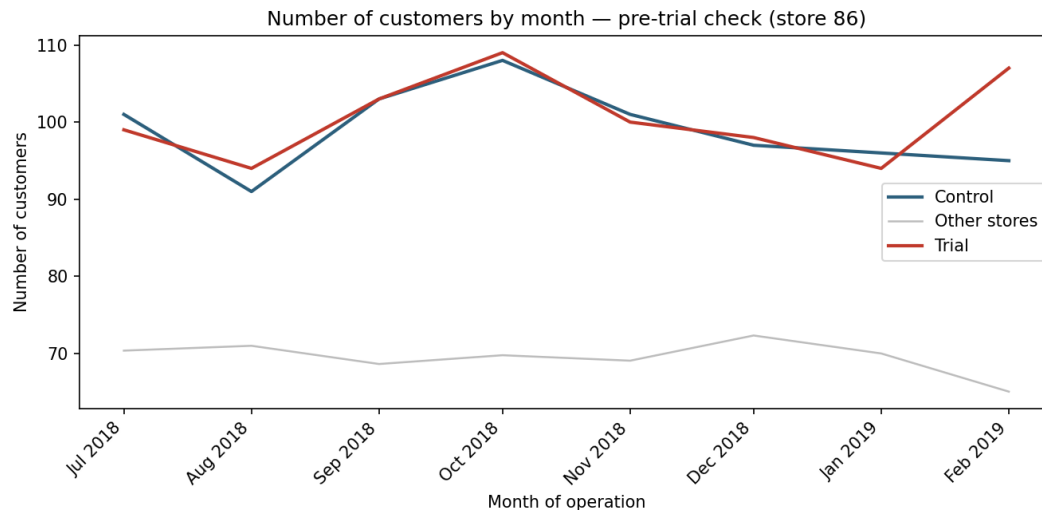
5. Trial Store 86

5.1 Pre-trial visual check

Before assessing the trial, we confirm trial store 86 and control store 155 moved together in the eight months before the trial (Jul 2018 – Feb 2019).



Pre-trial monthly total sales — store 86 vs control 155 vs other stores.



Pre-trial monthly customer counts — store 86 vs control 155 vs other stores.

The trial and control lines track each other closely prior to the trial, confirming this is a suitable control store.

5.2 Trial impact — total sales

```
scalingFactor = preTrial[trial].totSales.sum() / preTrial[control].totSales.sum()
scaledControlSales = control.totSales * scalingFactor
percentageDiff = abs(scaledControlSales - trial.totSales) / scaledControlSales
stdDev = percentageDiff[pre-trial months].std()
tValue = percentageDiff / stdDev # vs t-critical at 95th pct, df=7
```

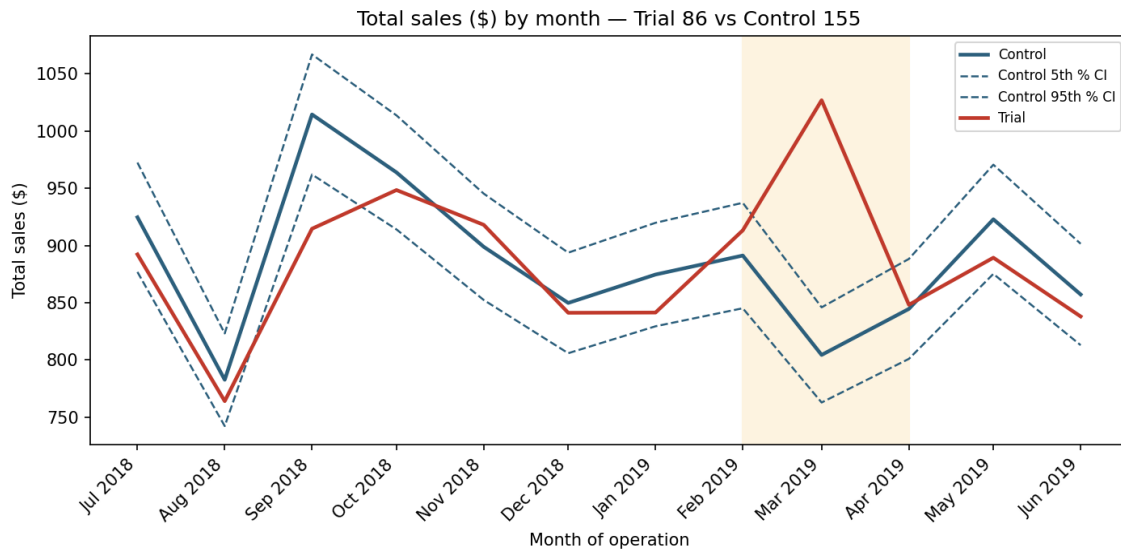


Figure: Trial vs control total sales with control store's 5th–95th percentile band (shaded = trial period).

Result: scaling factor = 0.970, pre-trial standard deviation of scaled %-difference = 0.026, t-critical (95th pct, df=7) = 1.89. Statistically significant sales uplift months: **02/2019, 03/2019**.

5.3 Trial impact — number of customers

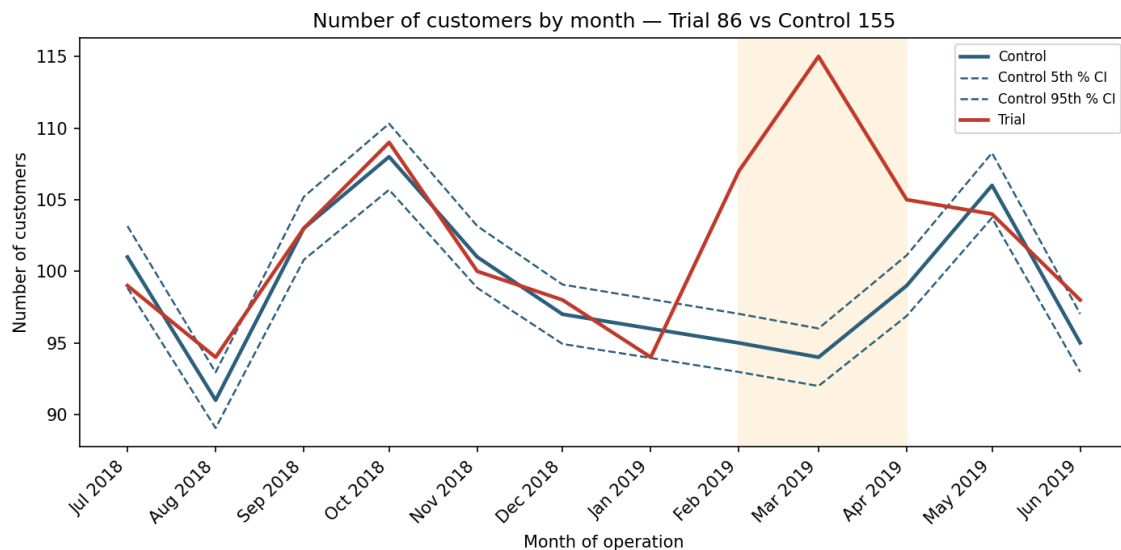


Figure: Trial vs control number of customers with control store's 5th–95th percentile band.

Result: statistically significant customer-count uplift months: **02/2019, 03/2019, 04/2019**.

5.4 What drove the change?

Store	Sales	Customers	Txn/Cust	Units/Txn	Price/Unit
Trial	6.3%	9.5%	-1.7%	0.8%	-2.1%
Control	-6.0%	-3.6%	-2.3%	1.1%	-1.3%

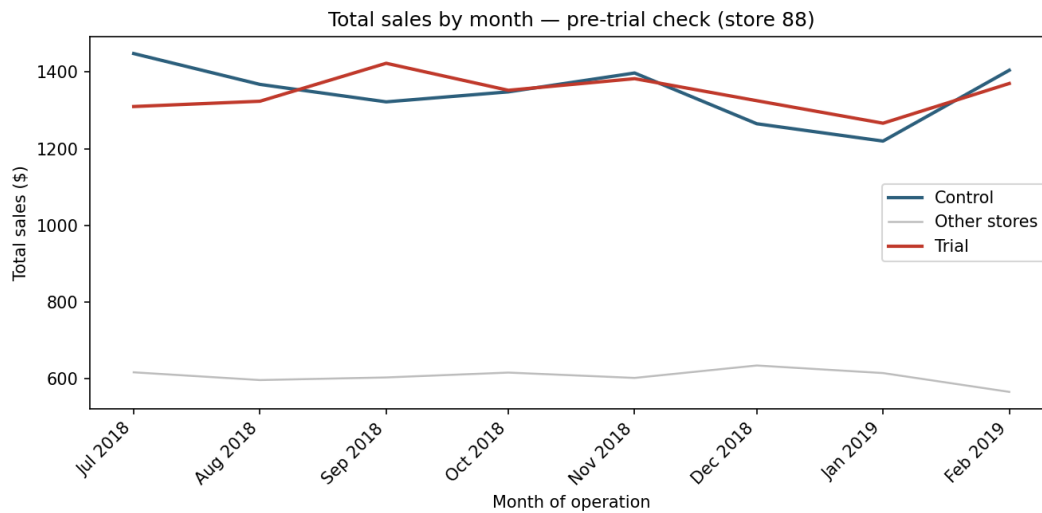
Average % change, trial period (Feb–Apr 2019) vs. pre-trial monthly average.

While control store 155's sales fell 6.0% over the same window (in line with the wider seasonal dip seen across most stores after the December peak), trial store 86's sales rose 6.3%. The gap is explained mainly by **more customers shopping at the store, rather than existing customers buying more per visit** (9.5% more customers vs. -1.7% change in visits/customer and 0.8% change in units/visit for the trial store).

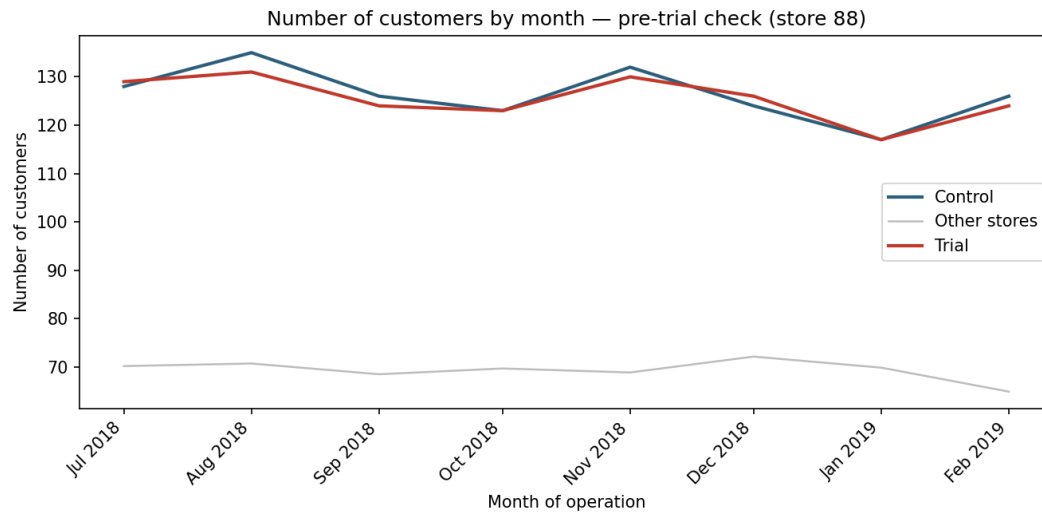
6. Trial Store 88

6.1 Pre-trial visual check

Before assessing the trial, we confirm trial store 88 and control store 237 moved together in the eight months before the trial (Jul 2018 – Feb 2019).



Pre-trial monthly total sales — store 88 vs control 237 vs other stores.



Pre-trial monthly customer counts — store 88 vs control 237 vs other stores.

The trial and control lines track each other closely prior to the trial, confirming this is a suitable control store.

6.2 Trial impact — total sales

```
scalingFactor = preTrial[trial].totSales.sum() / preTrial[control].totSales.sum()
scaledControlSales = control.totSales * scalingFactor
percentageDiff = abs(scaledControlSales - trial.totSales) / scaledControlSales
stdDev = percentageDiff[pre-trial months].std()
tValue = percentageDiff / stdDev # vs t-critical at 95th pct, df=7
```

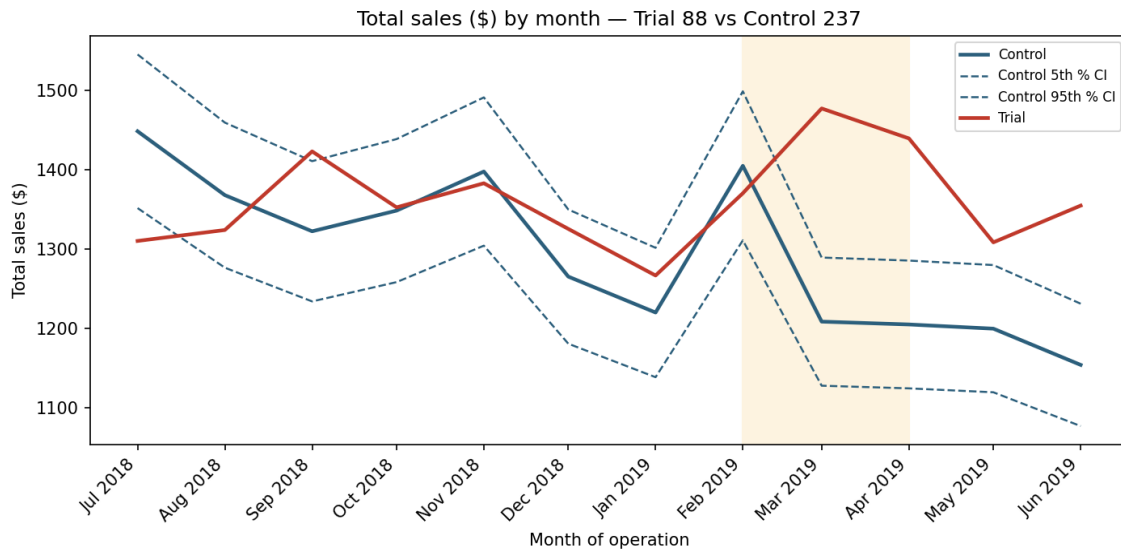


Figure: Trial vs control total sales with control store's 5th–95th percentile band (shaded = trial period).

Result: scaling factor = 1.002, pre-trial standard deviation of scaled %-difference = 0.033, t-critical (95th pct, df=7) = 1.89. Statistically significant sales uplift months: **03/2019, 04/2019**.

6.3 Trial impact — number of customers

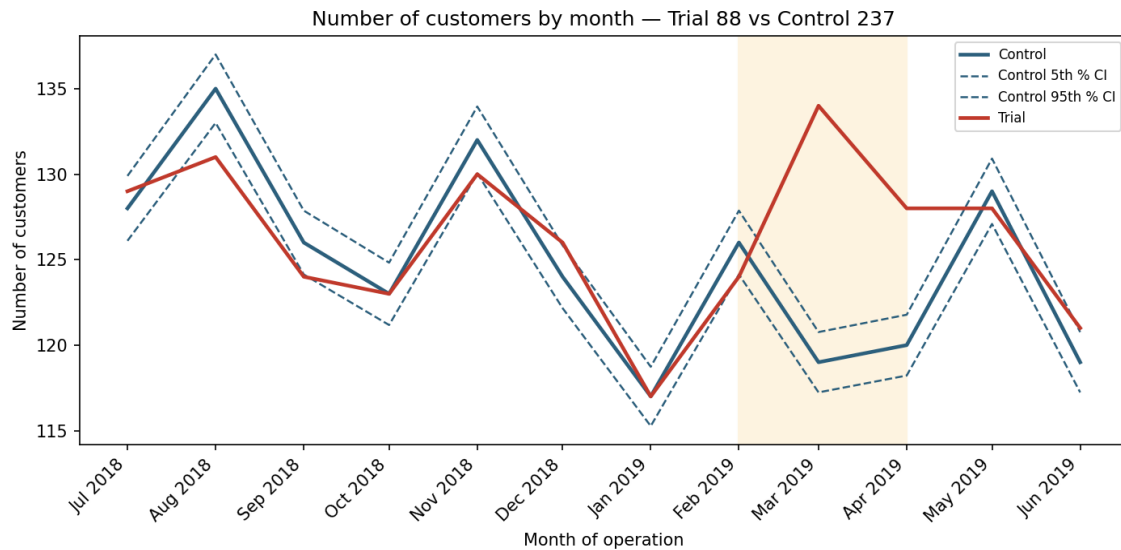


Figure: Trial vs control number of customers with control store's 5th–95th percentile band.

Result: statistically significant customer-count uplift months: **03/2019, 04/2019**.

6.4 What drove the change?

Store	Sales	Customers	Txn/Cust	Units/Txn	Price/Unit
Trial	6.6%	2.3%	2.9%	0.2%	1.0%
Control	-4.9%	-3.8%	-3.3%	1.5%	0.5%

Average % change, trial period (Feb–Apr 2019) vs. pre-trial monthly average.

While control store 237's sales fell 4.9% over the same window (in line with the wider seasonal dip seen across most stores after the December peak), trial store 88's sales rose 6.6%. The gap is explained mainly by **a mix of more customer visits and larger baskets per visit** (2.3% more customers vs. 2.9% change in visits/customer and 0.2% change in units/visit for the trial store).

7. Conclusion & Recommendations for Julia

Control stores 233, 155 and 237 were matched to trial stores 77, 86 and 88 respectively, based on pre-trial similarity in total sales and customer counts.

Trial store	Control store	Sig. sales months	Sig. customer months
77	233	2/3	2/3
86	155	2/3	3/3
88	237	2/3	2/3

Stores 77 and 88 show a statistically significant sales uplift in two of the three trial months (March and April), driven predominantly by an increase in the number of customers shopping at the store rather than by existing customers buying more chips per visit. **Store 86** also shows a significant sales uplift (February–March) and, notably, a significant increase in customer numbers across all three trial months — its sales effect is more muted than its customer effect, which is worth flagging to the Category Manager (e.g. to check whether promotional pricing in store 86 brought in more customers at a lower average spend).

Recommendations:

- Overall, the trial drove a significant, measurable increase in chip sales relative to matched control stores in two of the three trial stores, and a significant lift in customer traffic in all three — supporting a wider roll-out of the trial format.
- The uplift is being driven mainly by attracting more customers into the store, not by getting existing shoppers to buy more chips per visit. Roll-out plans and store-level KPIs should track footfall/new customers as the primary success metric, not just basket size.
- Follow up with the Category Manager on store 86 specifically — the strong customer growth without matching sales growth suggests promotional or pricing activity may be discounting the uplift; understanding this will help forecast the true sales impact of a full roll-out.
- Recommend extending the trial window or piloting in a second wave of stores with similar customer profiles to the three trial stores (see Task 1 segment work) to confirm the effect generalises before a full chain-wide roll-out.

— End of Task 2 initial findings —